

FRENY REJI

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EDUCATION

Indiana University, Bloomington <i>M.S., Data Science GPA: 3.53/4.0</i>	Aug 2024 – May 2026
Vellore Institute of Technology <i>Integrated MTech, Artificial Intelligence GPA: 8.8/10</i>	Jul 2019 – May 2024

TECHNICAL SKILLS

LLM & GenAI: Claude API, ReAct Agents, Retrieval-Augmented Generation (RAG), Multi-Agent Systems, Prompt Engineering, LLM-as-Judge Evaluation, Prompt-Injection Defense, LangGraph
Machine Learning: XGBoost, LSTM, Vision Transformers, CNNs, NLP, HDBSCAN, Statistical Modeling, End-to-End ML Pipelines, Time-series Analysis
Languages & Data: Python, SQL, R, Java, JavaScript, Unix Shell Scripting, HTML/CSS
Frameworks & Deployment: PyTorch, TensorFlow, Transformers, HuggingFace, scikit-learn, FastAPI, Flask, Streamlit, Docker, AWS, pytest, dbt, Airflow

EXPERIENCE

Machine Learning & Automation Intern Nokia Bengaluru, India	Aug 2023 – May 2024
- Reduced vulnerability analysis time by 60% by automating component-vulnerability matching across 3,000+ CVEs using HDBSCAN clustering and LLM-based entity linking, directly applicable to Telco KPI anomaly detection workflows. - Improved detection accuracy by 40% via transformer-based semantic similarity scoring, validating ML pipelines for production deployment. - Increased test coverage 35% and cut debugging time 25% by building optimized Robot Framework automation suites with CI/CD integration.	
Graduate Research Assistant – Data Visualization, Floating Solar Energy O'Neill School of Public & Environmental Affairs, IU Bloomington, IN	Jan 2026 – Present
- Selected for competitive \$3,000 fellowship: process satellite time-series data for 100+ global water bodies using ML workflows in Python and Google Earth Engine. - Build interactive dashboards integrating geospatial maps, ecological indicators, and trend analysis using JavaScript, SQL, Plotly, and Leaflet for interdisciplinary research stakeholders.	

PROJECTS

SourceTrace — Agentic Research Assistant GitHub Live Demo	2026
- Built an agentic RAG research assistant using Claude's tool-use API in a ReAct loop with cost-aware query routing, autonomously querying live NewsAPI and Wikipedia sources to produce cited, confidence-scored answers. - Designed and red-teamed a prompt-injection defense layer — untrusted-data wrapping plus pattern-based scanning — against adversarial retrieved content, validated by an automated adversarial test suite. - Built an LLM-judge evaluation harness comparing agentic vs. naive RAG across a 15-question set, showing a consistent relevance gain (+1.0 to +1.6 pts) across repeated runs.	
Sepsis Early Warning System GitHub Live Dashboard	2026
- Built an end-to-end clinical ML pipeline on synthetic ICU time-series: SOFA-based feature engineering (delta-vitals, rolling stats, shock index), XGBoost + LSTM ensemble with isotonic probability calibration, achieving 0.952 AUROC and 85.2% sensitivity with strict patient-level train/test splits to prevent leakage. - Designed a FastAPI backend and React/Recharts dashboard with a cohort explorer, per-patient risk timeline, and interactive alert-threshold tuner; deployed on Vercel with 10 alert episodes per 100 patient-days.	

ADDITIONAL PROJECTS

Smart Money Pulse — Behavioral Finance Stress Detection GitHub Live Dashboard	2026
Built a stress-detection pipeline in DuckDB (SQL window functions, 3-layer dbt DAG on Airflow) across 47K+ synthetic transactions, with a Streamlit dashboard surfacing a 4-tier cohort risk score.	
Hype vs. Box Office: Pre-Release Sentiment Analysis GitHub Live Dashboard	2025
Built an NLP pipeline (VADER + fine-tuned DistilBERT) over Reddit/TMDB data for 512 films, training XGBoost with SHAP to predict opening-weekend revenue (R^2: 0.71).	

CERTIFICATIONS & ACHIEVEMENTS

Amazon ML Summer School Participant | IBM AI Developer Associate | SQL Fundamentals (DataCamp)